

SECTION V: MATHEMATICAL FORMALIZATION

V.1 Overview

This section provides a **working mathematical framework** for Ψ ORM. It is not a complete, closed-form theory—it is a **scaffold**. The mathematics presented here is sufficient to:

1. **Define the core operators** and their relationships.
2. **Demonstrate the logical possibility** of consciousness-driven branch selection within standard quantum formalism.
3. **Generate specific, testable predictions** from the model.
4. **Provide a foundation** for future mathematical refinement.

The formalization is preliminary but not superficial. It is intentionally grounded in existing quantum mechanical structures, with modifications that encode the Ψ ORM ontology.

V.2 Operational Definition of Frequency (Ω)

For empirical purposes, the frequency operator Ω is proxied via:

- **EEG/MEG phase coherence (0–100 Hz)** — measuring neural oscillatory synchrony across cortical regions.
- **Microtubule dipole resonance** — a candidate biophysical substrate for quantum coherence in neural tissue.
- **Kuramoto order parameters** — a classical dynamical systems measure of phase synchronization across coupled oscillators.

The mathematical formalism of Ψ ORM remains scale-agnostic, meaning Ω can be mapped to any of these empirical proxies depending on the experimental context. Future work will refine the mapping as biophysical and quantum biological mechanisms are better understood. This operational flexibility allows Ψ ORM to generate testable predictions without committing to a specific biophysical substrate

prematurely.

V.3 Addressing Decoherence

Ψ ORM does not require literal quantum superposition of neural states. The frequency synchronization mechanism can be understood through either of two complementary frameworks:

Option A (Quantum Biology Grounding): Established quantum biological phenomena—such as proton tunneling in enzymes, topological protection in microtubules, or the Orch-OR coherence windows—provide plausible biophysical substrates for the frequency synchronization mechanism. If these mechanisms are confirmed, Ψ ORM's quantum formalism maps directly onto them.

Option B (Classical Dynamical Resonance): Alternatively, the frequency synchronization mechanism can be reframed as a *classical* dynamical resonance phenomenon—analogueous to phase-locking in coupled oscillators (Kuramoto models)—with quantum-inspired mathematics serving as a descriptive language rather than a literal physical claim. This interpretation is experimentally accessible via EEG/MEG phase coherence and does not require resolving decoherence challenges.

Ψ ORM is compatible with both interpretations. The mathematical formalism is a scaffold; the biophysical substrate remains an open empirical question. This positions Ψ ORM as a framework for investigating consciousness, not a definitive claim about its quantum nature.

V.4 The Conceptual-to-Mathematical Mapping

Conceptual Term	Mathematical Mapping	Operational Definition
Quantum Information Repository	Universal state vector $ \Psi\rangle$ in Hilbert space H .	The totality of all frozen moments. The soul.
Frozen Moments	Basis states $ x\rangle$ in the configuration space.	A specific configuration of reality at a given instant.
		The trajectory of

Consciousness Vector	Time-parameterized curve $ C(t)\rangle$ in projective Hilbert space.	experience through configuration space. The sequential illumination of frozen moments is the experience of time.
Frequency Operator	$\hat{\Omega} C\rangle = \Omega C\rangle$, where Ω is the phase coherence eigenvalue.	The degree of focus/coherence of the consciousness vector. Proxied via EEG/MEG phase coherence, microtubule resonance, or Kuramoto order parameters.
Biological Receiver	State $ B\rangle$ in the biological Hilbert space.	The physical substrate (brain/body) coupled to consciousness.
Frequency Synchronization	Interaction Hamiltonian $\hat{H}_{int} = g(\Delta\Omega) \cdot \hat{O}_C \otimes \hat{O}_B$.	The coupling between consciousness and the receiver.
Phase-Locking	Target state where $\Delta\Omega \rightarrow 0$ and $g(\Delta\Omega) \rightarrow g_{max}$.	The condition for maximal navigation efficiency.
Thought	Time-dependent control parameter $\theta(t)$ modulating $\hat{H}_{int}$.	The directed application of attentional weight. Generates thought-garbons.
Memory	The integrated structure of past experience; a memory-garbon.	Passive, past-oriented. Provides context and continuity.
Shift in Frequency	$\Delta\Omega = \langle C(t + dt) \hat{\Omega} C(t + dt) \rangle - \langle C(t) \hat{\Omega} C(t) \rangle$.	The change in coherence state driven by thought.
Bleed-Over	Non-zero entanglement entropy $S_E(V_A, V_B) > 0$.	Incomplete decoherence between adjacent vectors.
Bleed-Through	Non-zero overlap $ \psi_1\rangle \cdot \psi_2\rangle ^2 > 0$ for	Transient entanglement between previously

Bleed-through	$\ \langle V_A V_B \rangle\ ^2 > 0$ for $\Delta\Omega \approx 0$.	between previously orthogonal vectors.
Orthogonality Limit	$\ \langle V_A V_B \rangle\ ^2 = 0$ for $\ \Delta\Omega\ > T$.	The boundary beyond which vectors cannot interact.

V.5 The Qubit Toy Model (Strengthened)

Purpose: To demonstrate the logical possibility of consciousness-driven branch selection within standard quantum formalism.

Setup:

- Frozen moments: $|0\rangle$ and $|1\rangle$.
- Consciousness navigator: $|\uparrow\rangle$ and $|\downarrow\rangle$.
- Combined state: $|\Psi\rangle = \alpha|0\rangle|\uparrow\rangle + \beta|1\rangle|\downarrow\rangle$.

Total Hamiltonian:

$$H_{total} = H_C \otimes I + I \otimes H_B + H_{int}$$

Where:

- $H_C = \omega_C \hat{\sigma}_z^C$ (consciousness energy splitting)
- $H_B = \omega_B \hat{\sigma}_z^B$ (biological receiver energy splitting)
- $H_{int} = g(\Delta\Omega) \cdot \hat{\sigma}_x^C \otimes \hat{\sigma}_x^B$

The coupling strength g depends on the frequency mismatch $\Delta\Omega = |\omega_C - \omega_B|$. When $\Delta\Omega \rightarrow 0$ (resonance condition), $g(\Delta\Omega) \rightarrow g_{max}$.

Interaction Hamiltonian Expanded:

$$\hat{H}_{nav} = \omega \hat{\sigma}_z \otimes I + \theta(t) \hat{\sigma}_x \otimes \hat{\sigma}_x$$

Where:

- ω is the base frequency parameter.
- $\theta(t)$ is the "thought" parameter—a time-dependent control variable.

Schrödinger Equation:

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H}_{nav} |\Psi(t)\rangle$$

Solution (Resonance Condition):

For the initial state $|\Psi(0)\rangle = |0\rangle \otimes |\uparrow\rangle$, the transition probability to $|1\rangle \otimes |\downarrow\rangle$ is:

$$P(0 \rightarrow 1) = \frac{g^2}{\omega^2 + g^2} \sin^2 \left(\frac{\omega^2 + g^2}{\hbar} t \right)$$

Key Result: When g is maximized (resonance, $\Delta\Omega \rightarrow 0$), the transition probability approaches unity. This predicts maximal navigation efficiency at frequency alignment.

Connection to Experiment: The predicted transition probability should correlate with neural phase coherence measures (EEG PLV). Maximum phase locking corresponds to maximum g , which predicts higher navigation success rates.

V.6 Derivation of the Interaction Hamiltonian (Preliminary)

The interaction Hamiltonian is derived from the assumption that consciousness and the biological receiver are coupled quantum systems. The coupling is mediated by frequency synchronization:

1. **Assumption:** Consciousness and the biological receiver are quantum subsystems with distinct energy levels.
2. **Coupling:** The coupling strength g is a function of the frequency mismatch $\Delta\Omega$.
3. **Resonance:** When $\Delta\Omega \rightarrow 0$, the coupling is maximized, allowing coherent information transfer.
4. **Result:** The interaction Hamiltonian $\hat{H}_{int}$ encodes this coupling.

Formal Statement:

$$\hat{H}_{int}(\Delta\Omega) = g_0 \cdot \frac{\Gamma^2}{\Gamma^2 + (\Delta\Omega)^2} \cdot \hat{\sigma}_x^C \otimes \hat{\sigma}_x^B$$

Where:

- g_0 is the maximal coupling strength.

- Γ is the bandwidth of the resonance.
- $\Delta\Omega = |\hat{\Omega}_C - \hat{\Omega}_B|$.

This is a standard Lorentzian resonance function, adapted to the Ψ ORM ontology.

V.7 Integration with Existing MWI Mathematics

Ψ ORM is compatible with and extends several established mathematical frameworks:

Quantum Darwinism (Zurek):

- **What it provides:** Explains how pointer states are selected via environment-induced decoherence.
- **How Ψ ORM extends it:** Ψ ORM posits that branch selection is not purely environmental—it is mediated by frequency synchronization. The consciousness vector selects the branch that maximizes coherence with the biological receiver.

Everettian Decision Theory (Deutsch, Wallace):

- **What it provides:** Justifies the Born rule probabilities in MWI.
- **How Ψ ORM extends it:** Ψ ORM posits that the consciousness vector assigns weights to branches based on frequency similarity. The probability of experiencing a branch is proportional to the frequency alignment.

Self-Locating Uncertainty (Sebens and Carroll):

- **What it provides:** Addresses which branch the observer experiences.
- **How Ψ ORM extends it:** Ψ ORM defines the observer as the consciousness vector, and branch selection is driven by frequency synchronization.

V.8 Current Limitations and Future Directions

Current Limitations:

1. Reduced Hilbert space (2x2 states only).
2. Coupling constants unspecified for real biological systems.

3. No explicit connection to neural measurement protocols.
4. Not yet derived from first principles.

Future Work:

1. Expand to multi-dimensional configuration space.
2. Derive coupling constants from experimental data.
3. Map abstract operators to measurable physical quantities.
4. Develop a complete mathematical treatment of the model's predictions.

V.9 Falsifiability Statement (Mathematical)

The mathematical formalism generates specific predictions:

1. **Frequency Alignment Predicts Navigation Success:** The transition probability $P(0 \rightarrow 1)$ approaches unity when frequency alignment is achieved. This predicts that intentional, focused thought will show measurable frequency shifts ($\Delta\Omega$) and that these shifts will correlate with navigation success.
2. **Resonance Function Predicts Information Transfer:** The Lorentzian resonance function predicts that information transfer between consciousness vectors is maximized when $\Delta\Omega \rightarrow 0$. This predicts that inter-brain coherence studies will show high-frequency similarity during shared experiences.
3. **Orthogonality Limit Predicts Impossibility:** The boundary condition $\|\langle V_A | V_B \rangle\|^2 = 0$ for $\|\Delta\Omega\| > T$ predicts that communication between consciousnesses is strictly limited by frequency differences.

V.10 Summary

The mathematical formalization of Ψ ORM is:

1. **Grounded** in standard quantum formalism.
2. **Specific** in its predictions.
3. **Testable** via existing technology.
4. **Open** to refinement and extension.

It is a scaffold, not a cathedral. But it is a scaffold that points toward a cathedral.

This concludes Section V.